SET-H

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B.SC.[ECS] - II Sem III Practical Examination, October 2026

VSC1: Based on DSC1-3 and DSC1-4

Date & Time: Max Marks: 30

Instructions:

I) Attempt any **one** question from each section.

II) Each question carries **10 marks.**

III) The **Journal** and **Viva-Voce** carry **5 marks.**

Section-I: Based on DSC1-3 (Data Structures using C++)

Q.1 Write a program to find smallest and largest element of matrix.

Q.2 Write a program to implement binary search tree with tree traversal method.

Section-II: Based on DSC1-4 (Database Management Systems)

Q.3 Create a table student by applying proper constraints, insert at least 5 records and solve the following queries:

Field Name	Data Type	Constraint
Student_ID	INT	Primary Key
Student_Name	VARCHAR(50)	NOT NULL
Class	VARCHAR(20)	—
Gender	VARCHAR(10)	—
City	VARCHAR(30)	—
Marks	INT	—

a) Display the names and marks of all students.
b) Display students who have scored more than 60 marks.
c) Display students who belong to the city of Solapur.
d) Display students whose marks are between 50 and 80.
e) Display students whose names start with the letter 'A'.

Q.4 Create a table by applying proper constraints, insert at least 5 records and solve the following queries:

A] Supplier:

Field Name	Data Type	Constraint
Supplier_ID	INT	Primary Key
Supplier_Name	VARCHAR(50)	NOT NULL
City	VARCHAR(30)	—

B] Product:

Field Name	Data Type	Constraint
Product_ID	INT	Primary Key
Product_Name	VARCHAR(50)	NOT NULL
Price	DECIMAL(10,2)	—
Quantity	INT	—
Supplier_ID	INT	Foreign Key referencing Supplier(Supplier_ID)

a) Display product names, prices, and supplier names using an `INNER JOIN`.
b) Display all suppliers, including those who supply no products, using a `LEFT JOIN`.
c) Display products costing more than ₹1,000 with their supplier names.
d) Display products whose prices are greater than the average product price.
e) Display products supplied by the supplier named 'ABC Traders'.